med6m: a LoRA adapter for Qwen3-8B on medical multiple-choice QA

Sealed-test evidence for checkpoint 443 at LoRA scale 0.5

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Hugging Face `odelpa/qwen3-8b-med6m` · GitHub `usergdp136/qwen3-8b-med6m`

Abstract

We release a LoRA adapter (rank 64) for `unsloth/Qwen3-8B-unsloth-bnb-4bit`, trained with QLoRA on a six-month medical mix (MedMCQA-train, MedQA-train, Italian and English medical Wikipedia) plus 22% general replay. On a *sealed, single-use* test of 3 766 medical multiple-choice items (MedMCQA, MedQA, six MMLU medical subdomains, PubMedQA) the adapter at LoRA scale $s = 0.5$ improves accuracy over the base model by +2.34 percentage points (pp), 95% paired-bootstrap CI $[+1.14, +3.53]$, exact McNemar $p = 1.4 \times 10^{-4}$ (306 items only the adapter gets right vs. 218 only the base). The gain is concentrated in MedMCQA (+2.89 pp) and PubMedQA (+2.78 pp); MedQA and MMLU-med are positive but their CIs include zero; the Italian general guard (ARC-it) shows no regression (+0.74 pp, CI $[-1.29, +2.76]$). Important limits: the evaluated artefact is checkpoint 443, not the final 886; evaluation is answer-letter log-probability, not free generation; no clinical validity is claimed.

1 Method

Model and adapter. Base: Qwen3-8B (Apache-2.0), loaded in 4-bit NF4 (bitsandbytes). The adapter is a LoRA on all attention and MLP projections (`q,k,v,o,gate,up,down`), rank $r = 64$, $\alpha = 64$, no dropout. For a frozen weight W_0 the adapted weight is

$$W \;=\; W_0 \;+\; s\,\frac{\alpha}{r}\,BA, \qquad A \in \mathbb{R}^{r \times d_{\text{in}}},\; B \in \mathbb{R}^{d_{\text{out}} \times r}, \tag{1}$$

where s is the inference-time LoRA scale (here $s = 0.5$, $\alpha/r = 1$): the learned update is applied at half strength.

Training. Two consecutive QLoRA stages on a Tesla T4: stage `S1_v2` (885 steps, lr 5×10^{-5}, seed 3407) on the four medical sources only; stage `S1_v2r` (planned 886 steps, lr 1.3×10^{-5}, warm-up 25, seed 3408, grad. accumulation 16, max length 2048), initialised from the last adapter of stage 1 and adding 22% replay (OASST2, Aya, Dolly, Italian Wikipedia, FineWeb-Edu, FineWeb-2-it) to limit forgetting. The **published artefact is checkpoint 443 of stage `S1_v2r`** (50% of the stage). About 29.5M effective target tokens were planned for the second stage (medical 78%, replay 22%).

2 Evaluation protocol

Every item is scored by the log-probability of the answer letter. With the chat template (thinking disabled) and the prompt truncated to 2048 tokens, the prediction is

$$\hat{y} \;=\; \arg\max_{\ell \in \mathcal{L}} \; z_\ell, \tag{2}$$

where z_ℓ is the next-token logit of letter ℓ (with and without leading space) and $\mathcal{L}$ the option letters. Free-form generation is *not* evaluated. Benchmarks: MedMCQA (validation), MedQA-USMLE-4-options (test), six medical MMLU subdomains, PubMedQA (`pqa_labeled`, question only, no context), and ARC-Italian (challenge) as general guard. The dev/test split is deterministic: SHA-1 of `"split|"+ID`, parity of the digest. Training data were decontaminated against dev and test with 10-gram matching. The sealed test was run *once* (marker `TEST_USED`, 1 Nov 2026) for the base and for checkpoint 443 at $s \in \{0.5, 0.75\}$; the test is therefore consumed and cannot be reused for further selection.

3 Statistics

For item i let $b_i, c_i \in \{0, 1\}$ be the correctness of base and adapter and $d_i = c_i - b_i \in \{-1, 0, 1\}$. The reported effect is

$$\Delta_{\text{pp}} \;=\; \frac{100}{n} \sum_{i=1}^{n} d_i \;=\; \frac{100\,(n_+ - n_-)}{n}, \tag{3}$$

with n_+ (n_-) the items only the adapter (only the base) answers correctly. Confidence intervals are percentile paired bootstraps, implemented as multinomial resampling of the counts,

$$\begin{aligned} (n_-^*, n_0^*, n_+^*) &\sim \text{Mult}\big(n;\, \hat{p}\big), \quad \hat{p} = \big(\tfrac{n_-}{n}, \tfrac{n_0}{n}, \tfrac{n_+}{n}\big) \\ \text{CI}_{95} &= \big[q_{0.025},\, q_{0.975}\big]\big\{100\,\tfrac{n_+^* - n_-^*}{n}\big\}, \end{aligned} \tag{4}$$

with $B = 20\,000$ replicates and seed 20261101. The exact two-sided McNemar test on the discordant pairs ($n_- + n_+ = m$) is

$$p \;=\; \min\Big\{1,\; 2 \sum_{k=0}^{\min(n_-, n_+)} \binom{m}{k} 2^{-m}\Big\}. \tag{5}$$

The pool concatenates the four medical benchmarks and weights *items*, not benchmarks. No correction for multiple comparisons is applied; per-benchmark p-values are descriptive.

4 Results on the sealed test

Table 1 gives the primary result (scale 0.5); Figure 1 shows the differences with CIs for both scales; Figure 2 the accuracies and Figure 3 the gains and losses per item.

Reading. The medical pool improves by +2.34 pp (CI excludes 0, $p = 1.4 \times 10^{-4}$). Two benchmarks drive it: MedMCQA (the training mix contains MedMCQA-train, so part of the gain is in-distribution adaptation to the exam format) and PubMedQA. MedQA (+1.55) and MMLU-med (+0.74) have CIs that include zero: they are *not* confirmed improvements. The guard ARC-it shows no regression on the sealed test. Scales 0.5 and 0.75 are statistically indistinguishable on the pool (+2.34 vs. +2.44 pp).

Table 1: Sealed test, checkpoint 443, **scale 0.5** vs. base. Accuracy in %; "+/−" = items only the adapter / only the base answer correctly. Δ in percentage points with 95% bootstrap CI; exact McNemar p.

Set	n	Base	Adapter	Δ (pp)	95% CI	+/−	McNemar p
MedMCQA	2078	57.31	60.20	+2.89	[+1.15, +4.62]	200/140	0.001
MedQA	646	62.85	64.40	+1.55	[−1.55, +4.64]	57/47	0.378
MMLU-med	538	78.25	79.00	+0.74	[−1.67, +3.16]	25/21	0.659
PubMedQA	504	57.94	60.71	+2.78	[+0.60, +5.16]	24/10	0.024
Medical pool (n=3766)	3766	61.34	63.67	+2.34	[+1.14, +3.53]	306/218	<0.001
ARC-it (guard)	543	88.21	88.95	+0.74	[−1.29, +2.76]	17/13	0.585

Table 2: Same test, scale 0.75 (reported for transparency; both scales were evaluated on the same consumed test).

Set	n	Base	Adapter	Δ (pp)	95% CI	+/−	McNemar p
MedMCQA	2078	57.31	60.97	+3.66	[+1.83, +5.49]	230/154	<0.001
MedQA	646	62.85	63.31	+0.46	[−2.63, +3.72]	56/53	0.848
MMLU-med	538	78.25	79.00	+0.74	[−1.86, +3.35]	28/24	0.678
PubMedQA	504	57.94	59.72	+1.79	[−0.40, +4.17]	22/13	0.175
Medical pool (n=3766)	3766	61.34	63.78	+2.44	[+1.17, +3.69]	336/244	<0.001
ARC-it (guard)	543	88.21	88.77	+0.55	[−1.66, +2.76]	20/17	0.743

5 Final checkpoint 886 on dev (not independent)

The final checkpoint was evaluated on the dev split only, which guided training decisions (gate, scale). It is therefore *not* an independent confirmation and is not the published artefact. On

Table 3: DEV split, final checkpoint 886 at scale 1.0 vs. base (pool $n = 3779$).

Set	n	Base	Final	Δ (pp)	95% CI	+/−
MedMCQA	2105	56.25	59.48	+3.23	[+1.38, +5.08]	238/170
MedQA	627	62.04	65.87	+3.83	[+0.48, +7.18]	70/46
MMLU-med	551	77.13	76.41	−0.73	[−3.63, +2.18]	30/34
PubMedQA	496	53.83	55.44	+1.61	[−0.81, +4.23]	25/17
Medical pool	3779	59.94	62.48	+2.54	[+1.22, +3.84]	363/267
ARC-it (guard)	589	86.42	84.55	−1.87	[−4.07, +0.34]	18/29

dev the final checkpoint gains +2.54 pp on the medical pool but loses −0.73 pp on MMLU-med and −1.87 pp on ARC-it (CIs include 0). The historical gate (two standard errors) accepted it; a stricter criterion (margin −0.5 pp, ARC-it floor 85.42%) would reject it. The gate criterion was revised on 31 Oct 2026 and the 50% checkpoint was re-rated post hoc. For these reasons we publish checkpoint 443 (which was tested on the sealed set) and *not* the final one.

6 Limitations and risks

- **Checkpoint vs. final.** Evidence concerns checkpoint 443; nothing is claimed for 886.
- **Post-selection.** Two scales were evaluated on the same consumed test; scale 0.5 was preferred from dev criteria (smaller ARC-it change, larger PubMedQA gain), but a pre-registered choice does not exist.

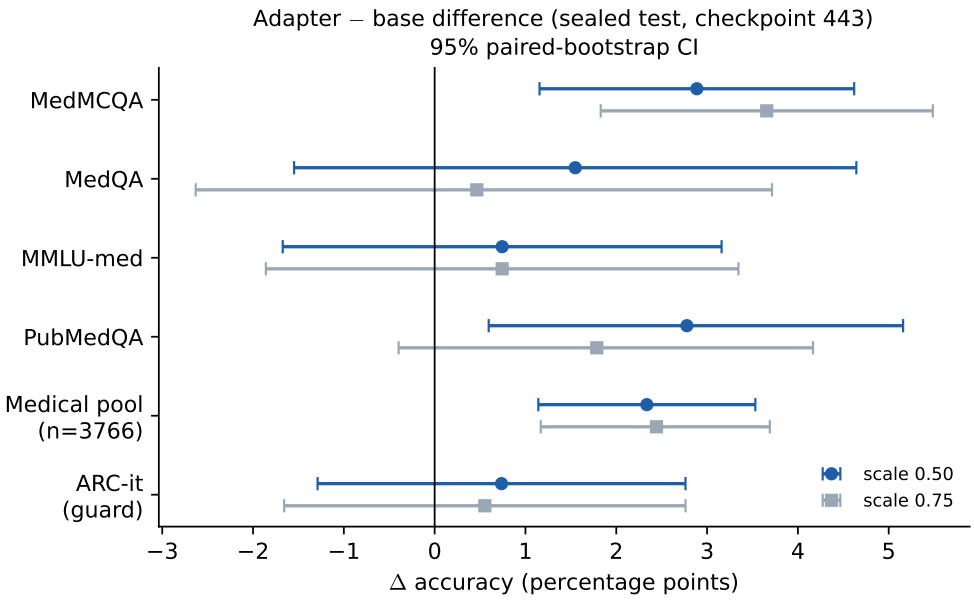


Figure 1: Adapter minus base, sealed test. Points: Δ_{pp} (Eq. 3); bars: 95% CI (Eq. 4).

- **No multiplicity correction**; benchmark-level CIs are descriptive. MedQA and MMLU-med are not significant.
- **Task type.** Multiple-choice letter log-probability only; no free-text answers, no safety evaluation, no clinical validation. *Do not use for medical advice.*
- **General capability checks pending:** GSM8K and IFEval on checkpoint 443 have *not* yet been run; only the Italian ARC guard is reported.
- **Provenance.** MedMCQA-train is declared Apache-2.0 but its exam-question provenance is not independently verified; dataset/model revisions were not pinned at training time (we report the cached revisions available on the training machine, Appendix A).
- **Network audit.** A static review of the training/evaluation code found no non-Hugging-Face endpoints and `report_to='none'`; a dynamic (traffic) audit was not performed retroactively (Appendix B).

7 Licences and attribution

Base model: Qwen3-8B, Apache-2.0 (© Qwen Team / Alibaba Cloud; 4-bit repack by Unsloth). Training sources and licences: MedMCQA-train (Apache-2.0, declared), MedQA-train (MIT, via the `openlifescienceai/medqa` train split), Wikipedia IT/EN (CC BY-SA 3.0 / GFDL), Dolly-15k (CC BY-SA 3.0), OASST2 and Aya (Apache-2.0), FineWeb-Edu and FineWeb-2 (ODC-By 1.0). All are licences that permit commercial use; none is non-commercial. Because several sources are share-alike, the adapter, the article and the figures are released under **CC BY-SA 4.0** and the code under Apache-2.0; attribution is in `ATTRIBUTION.md`. Whether share-alike legally propagates to model weights is unsettled; we apply it conservatively.

8 Reproducibility

`code/recompute.py` recomputes every number in the tables from the paired counts (CPU, NumPy, no network); `code/make_figures.py` regenerates the figures. The Codex-assisted audit and our

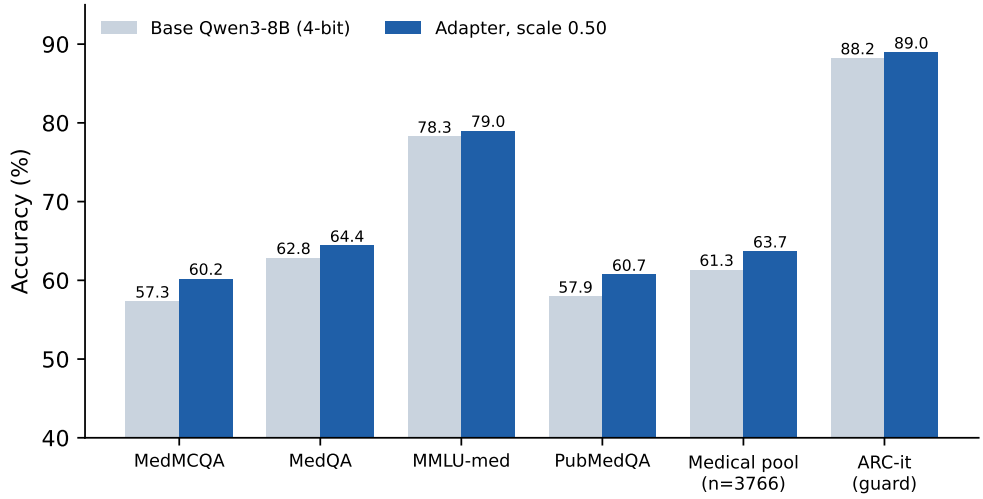


Figure 2: Accuracy of the base and of the adapter (scale 0.5).

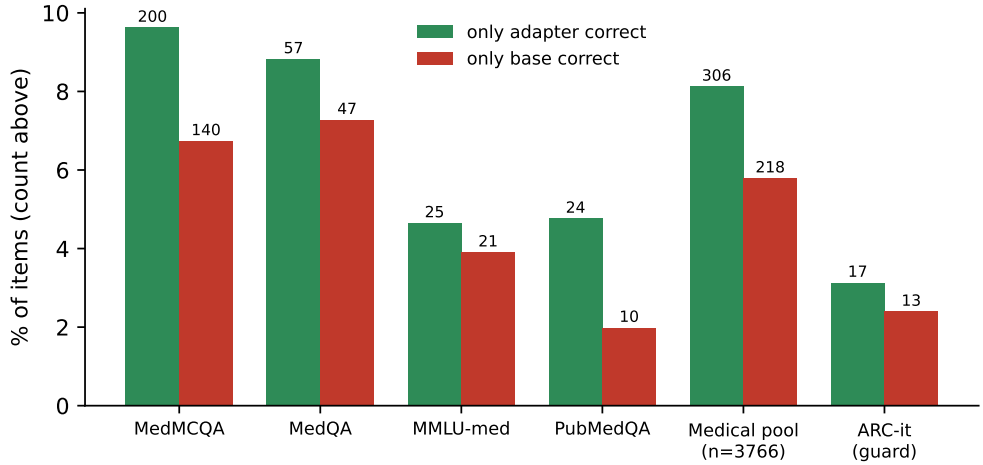


Figure 3: Items only the adapter gets right (n_+) versus only the base (n_-).

recomputation agree to the second decimal (pool +2.3367, CI $[1.1418, 3.5316]$). Training and evaluation scripts are provided for inspection; they need the original environment (Unsloth 2026.10.7, Transformers 5.17.0, PEFT 0.21.2, TRL 1.15.0, bitsandbytes 0.50.2, PyTorch 2.11.0+cu128).

A Pinned artefacts

B Network audit note

Scope: the med6m code (`build_s1v2r.py`, `train_stage_r.py`, `eval_mc.py`, `s0_data.py`, `s0b_data.py`). Finding: network access is limited to Hugging Face datasets/hub downloads for data preparation; `report_to='none'` (no W&B or other trackers); training and scoring read local JSONL files. `pmc.py` (Europe PMC S3) belongs to the data-preparation helpers and is not part of training/evaluation. Not verified: environment flags (`HF_HUB_OFFLINE`, `UNSLOTH_DISABLE_STATISTICS`) were not found in the launch scripts, so library telemetry cannot be excluded; a clean offline re-run with traffic capture is recommended.

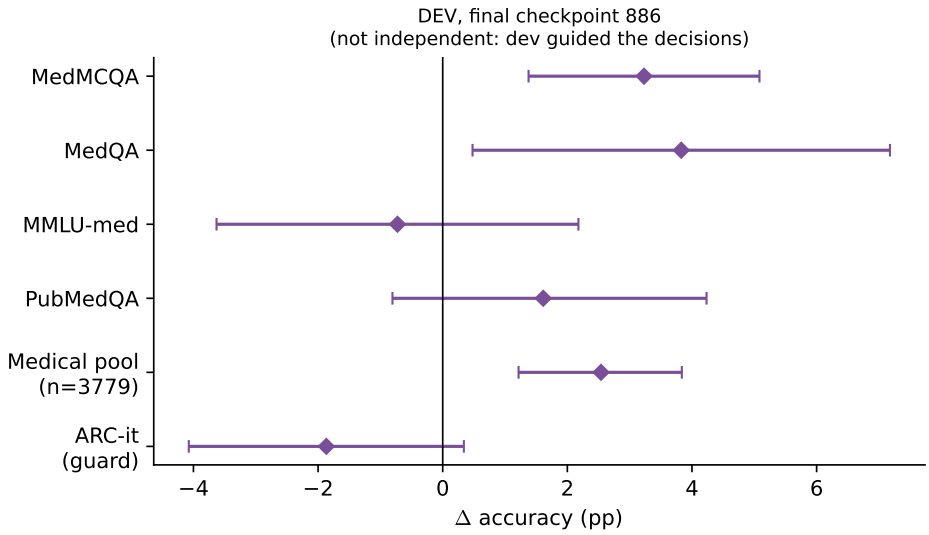


Figure 4: Dev differences for the final checkpoint 886.

Artefact	SHA-256 / revision
`adapter_model.safetensors`	`0dfdebe51f325a0d0e1cc5851193a03c4d6aef898ba3b3a595145ab508c4a5bf`
`adapter_config.json`	`dcabc2e1d6661d1ad5a18f475a7fd08c875e8cba8664ffbea95bff32d0ce6943`
`tokenizer.json`	`476870a1f2fb6f6a2759a6ede2383bf9d5d738f17844563b65c91965b722ae09`
Base `unsloth/Qwen3-8B-unsloth-bnb-4bit`	revision `62efd7f9d748e394734a7adae2adf96e13a2abc8` (cached 24 Oct 2026)
`openlifescienceai/medmcqa`	`91c6572c454088bf71b679ad90aa8dffcd0d5868`
`GBaker/MedQA-USMLE-4-options`	`0fb93dd23a7339b6dcd27e241cb9b5eca62d4d18`
`cais/mmlu`	`c30699e8356da336a370243923dbaf21066bb9fe`
`qiaojin/PubMedQA`	`9001f2853fb87cab8d220904e0de81ac6973b318`
`sapienzanlp/arc_italian`	`1860d09fba59b9b2576335ab472346f63c1becc8`
`wikimedia/wikipedia`	`b04c8d1ceb2f5cd4588862100d08de323dccfbaa`

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